INTRODUCTION TO INFORMATION GEOMETRY

Square-Root Coordinates and the Sphere

Week 8 · Lecture 1

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A new coordinate for each probability

For an interior categorical law $P = (p_i)$, define

$$s_i = 2\sqrt{p_i}.$$

Since $\sum_i p_i = 1$,

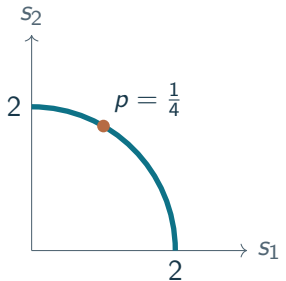
$$\sum_i s_i^2 = 4.$$

The image lies in the positive orthant of a sphere of radius 2.

Bernoulli laws form a quarter-circle

For $P = (p, 1-p)$, the image is

$$s(p) = (2\sqrt{p}, 2\sqrt{1-p}).$$



Mapping a Bernoulli law

Map $P = (\frac{1}{4}, \frac{3}{4})$ to square-root coordinates.

$$s = (2\sqrt{\tfrac{1}{4}}, 2\sqrt{\tfrac{3}{4}}) = (\ ?,\ ?).$$

Check that $\|s\|^2 = 4$.

Mapping a Bernoulli law

Map $P = (\frac{1}{4}, \frac{3}{4})$ to square-root coordinates.

$$s = (1, \sqrt{3}).$$

Indeed $\|s\|^2 = 1 + 3 = 4$.

Square roots preserve Fisher length

Differentiate $s_i = 2\sqrt{p_i}$:

$$ds_i = \frac{dp_i}{\sqrt{p_i}}.$$

Therefore the Euclidean squared length on the sphere is

$$\sum_i ds_i^2 = \sum_i \frac{dp_i^2}{p_i} = ds_F^2.$$

The square-root map is an isometry between the positive simplex and the positive spherical orthant.

The shortest spherical route

For two non-antipodal points on a sphere, the shortest path is the shorter arc of the great circle through them. On a radius-2 sphere, its length is twice the central angle:

$$d = 2\alpha.$$

Square-root images of probability laws lie in one positive orthant; their shorter arc remains inside it.

The Bhattacharyya overlap

For two categorical laws $P = (p_i)$ and $R = (r_i)$, define

$$B(P, R) = \sum_i \sqrt{p_i r_i}.$$

Their square-root vectors have dot product $4B(P, R)$, hence

$$\cos \alpha = B(P, R).$$

This determines Fisher distance directly from the probabilities.

Fisher distance for categorical laws

$$d_F(P, R) = 2 \arccos\left(\sum_i \sqrt{p_i r_i}\right)$$

The expression is symmetric and measures shortest-path length.

An exact distance from three probabilities

Let $P = (.5, .3, .2)$ and $R = (.2, .5, .3)$. Calculate the overlap

$$B(P, R) = \sqrt{.10} + \sqrt{.15} + \sqrt{.06}.$$

Then determine $d_F(P, R)$.

An exact distance from three probabilities

Let $P = (.5, .3, .2)$ and $R = (.2, .5, .3)$. The overlap is $B \approx .948475$.

$$d_F(P, R) = 2 \arccos B \approx .644817.$$

Boundary distributions have finite distance

For disjoint pure outcomes, such as $P = (1, 0)$ and $R = (0, 1)$,

$$B(P, R) = 0.$$

Thus the limiting Fisher distance is

$$d_F(P, R) = \pi.$$

The open positive simplex can be completed by adding its boundary at finite distance.

KL divergence and Fisher distance

KL divergence compares expected log loss and depends on direction. Fisher distance is symmetric because it is intrinsic shortest-path length. Both arise from the same local quadratic information structure.